Facultad de Ciencias

Universidad Autónoma de México
Electromagnetismo II – Examen 1
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Nota. Quiero ver si me aceptan mis trabajos en ingles jaja, para acostumbrarme y aprender.

1.- Problema: (30pts) Calcula la fuerza sobre la carga $+2q$ de la figura que se muestra a continuación. El plano XY representa un plano conductor aterrizado.

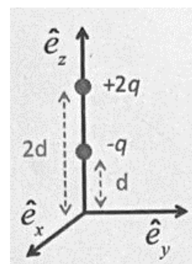
We can calculate the force on the load using the image method and adding all the interacting forces, by superposition principle. We can calculate the force using Coulomb, and it is also restricted to only the axis $\hat{e}_z$.

$$\bar{F} = \bar{F}_1 + \bar{F}_2 + \bar{F}_3 = \frac{1}{4\pi\epsilon_0}\frac{(2q)(-q)}{d^2}\hat{e}_z + \frac{1}{4\pi\epsilon_0}\frac{(2q)(q)}{(3d)^2}\hat{e}_z + \frac{1}{4\pi\epsilon_0}\frac{(2q)(-2q)}{(4d)^2}\hat{e}_z$$

$$= \frac{2q^2}{4\pi\epsilon_0}\left(-\frac{1}{d^2} + \frac{1}{9d^2} - \frac{2}{16d^2}\right)\hat{e}_z = \frac{2q^2}{4\pi\epsilon_0 d^2}\left(-1 + \frac{1}{9} - \frac{2}{16}\right)\hat{e}_z$$

$$= \frac{2q^2}{4\pi\epsilon_0 d^2}\left(-1 + \frac{1}{9} - \frac{1}{8}\right)\hat{e}_z = \frac{2q^2}{4\pi\epsilon_0 d^2}\left(-\frac{73}{72}\right)$$

$$= -\frac{q^2}{4\pi\epsilon_0 d^2}\left(\frac{73}{36}\right)\hat{e}_z$$



2. Problema: (30pts) Una línea infinita con densidad lineal de carga uniforme $-\lambda$ se coloca a una distancia a sobre un plano conductor aterrizado. La línea es paralela al eje y y el plano conductor está localizado en el plano XY.

(a) Calcula el potencial en la región superior del plano conductor.

(b) Calcula la densidad de carga inducida en el plano conductor.

To calculate the potential, we must use the imaging method. Already in class and in an assignment, we arrive at the potential of an infinite line with uniform charge density is given by:

$$\phi(\bar{r}) = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{|\bar{r} - \bar{r}'|}{r_0}$$

So, by superposition principle, applied to the problem where the second loading line is generated by the image method:

$$\phi(x, z) = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{|\bar{r} - \bar{r}'|}{r_0} - \frac{\lambda}{2\pi\epsilon_0}\ln\frac{|\bar{r} - \bar{r}''|}{r_0}$$

Being $\bar{r}' = a\hat{e}_z$ y $\bar{r}'' = -a\hat{e}_z$, this by construction, since they are each at a distance a.

$$\phi(x,z) = \frac{\lambda}{2\pi\epsilon_0}\left(\ln\frac{|\bar{r} - a\hat{e}_z|}{r_0} - \ln\frac{|\bar{r} + a\hat{e}_z|}{r_0}\right) = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{\frac{|\bar{r} - a\hat{e}_z|}{r_0}}{\frac{|\bar{r} + a\hat{e}_z|}{r_0}}\right) = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{|\bar{r} - a\hat{e}_z|}{|\bar{r} + a\hat{e}_z|}\right)$$

We will calculate$|\bar{r} \pm a\hat{e}_z|$, we will use Pythagoras in the construction, remember that in this case it is parallel to and: $|\bar{r} \pm a\hat{e}_z| = \sqrt{x^2 + (z \pm a)^2}$

Therefore,

$$\phi(x,z) = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{\sqrt{x^2 + (z-a)^2}}{\sqrt{x^2 + (z+a)^2}}\right) = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\left(\frac{x^2 + (z-a)^2}{x^2 + (z-a)^2}\right)^{\frac{1}{2}}\right) = \frac{\lambda}{4\pi\epsilon_0}\ln\left(\frac{x^2 + (z-a)^2}{x^2 + (z-a)^2}\right)$$

We can calculate the induced charge density with the following equation:

$$\begin{aligned}
\sigma_i(x) &= -\epsilon_0\frac{\partial\phi}{\partial z}\bigg|_{z=0} = -\epsilon_0\frac{\lambda}{4\pi\epsilon_0}\left[\frac{2(z-a)}{x^2 + (z-a)^2} - \frac{2(z+a)}{x^2 + (z+a)^2}\right]\bigg|_{z=0} \\
&= -\frac{\lambda}{4\pi}\left[\frac{2(0-a)}{x^2 + (0-a)^2} - \frac{2(0+a)}{x^2 + (0+a)^2}\right] = -\frac{\lambda}{4\pi}\left[-\frac{2a}{x^2 + a^2} - \frac{2a}{x^2 + a^2}\right] \\
&= -\frac{\lambda}{4\pi}\left[-\frac{2a}{x^2 + a^2} - \frac{2a}{x^2 + a^2}\right] = \frac{\lambda}{4\pi}\left(\frac{4a}{x^2 + a^2}\right) = \frac{\lambda a}{\pi(x^2 + a^2)}
\end{aligned}$$

3. Problema: (40pts) La superficie de una esfera conductora de radio R, centrada en el origen, se divide en dos partes por el plano YZ. La parte hacia el eje positivo $\hat{e}_x$ se mantiene a un potencial constante ϕ_0 y la parte hacia $-\hat{e}_x$ está a potencial $-\phi_0$. Calcula el potencial como una representación en serie (en base esférica) para todos los puntos interiores a la esfera.

Hint: Recuerda que $P_n^m(x) = (1-x^2)^{m/2}\frac{d^m}{dx^m}P_n(x)$ y que $P_n^{-m} = (-1)^m\frac{(n-m)!}{(n+m)!}P_n^m(x)$. También recuerda que $Y_l^m(\theta,\varphi) = (-1)^m\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(-+m)!}}P_l^m(\cos\theta)e^{im\varphi}$.

The solution in spherical coordinates presented in the problem is based on solving the Laplace equation $\nabla^2\phi = 0$ for an electric potential ϕ inside a conducting sphere of radius R, centered on the origin. Since there are no charges inside the sphere, the potential must satisfy Laplace's equation. Furthermore, in this case, the potential must be finite, so we do not take into account the terms $B_{l,m}r^{-(l+1)}$, Well, as was said, when $r \to 0$. Then we have that the potential is given by:

$$\phi(r,\theta,\varphi) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} A_{l,m}r^l Y_{l,m}(\theta,\varphi)$$

The problem asks us how the sphere is divided, so that it is half and half, we can put the following correspondence rule for the potential.

$$\phi(R,\theta,\varphi)\begin{cases} -\phi_0, & si\ \varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \phi_0, & si\ \varphi \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right] \end{cases}$$

From the orthogonality of the harmonics we will consider the following expression that gives us the value for the coefficients $A_{l',m'}R^{l'}$ (we already evaluated in $r = R$, to satisfy the boundary conditions), and solving for the constants.

$$A_{l',m'}R^{l'} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} A_{l,m}R^{l} \int_{0}^{2\pi} \int_{-1}^{1} d\varphi \, Y_{l,m} Y^{*}_{l',m'} d(cos\theta)$$

In this way, using the Hint and changing the variable, $cos\theta = x$,

$$A_{l,m} = \frac{1}{R^l}\sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} \int_{0}^{2\pi} \left(\sum_{l=0}^{\infty} \sum_{m=-l}^{l} A_{l,m}R^{l} Y_{l,m}(\theta,\varphi) \right) e^{-im\varphi} d\varphi \int_{-1}^{1} P_l^m(x) dx$$

Let's analyze the integral

$$\int_{0}^{2\pi} \left(\sum_{l=0}^{\infty} \sum_{m=-l}^{l} A_{l,m}R^{l} Y_{l,m}(\theta,\varphi) \right) e^{-im\varphi} d\varphi = \int_{0}^{2\pi} \phi e^{-im\varphi} d\varphi$$

$$= \frac{i\phi_0}{m} e^{-im\varphi} \Bigg|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \frac{i\phi_0}{m} e^{-im\varphi} \Bigg|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} = 2\frac{i\phi_0}{m} e^{-im\varphi}\left(1 - e^{-im\pi}\right)$$

Let's analyze what happens for the values of m,

For $m = 0$ or pair:

$$2\frac{i\phi_0}{m} e^{-im\varphi}(1-1) = 0$$

For m odd:

$$2\frac{i\phi_0}{m} e^{-im\varphi}(1+1) = 4\frac{i\phi_0}{m} e^{-im\varphi}$$

Therefore,

$$A_{l,m} = \frac{1}{R^l}\sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} 4\frac{i\phi_0}{m} e^{-im\varphi} d\varphi \int_{-1}^{1} P_l^m(x) dx$$

For the odd m's,

Finally, for the interior points of the sphere, with odd, another case is 0:

$$\phi(r,\theta,\varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{R^l}\sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} 4\frac{i\phi_0}{m} e^{-im\varphi} d\varphi \int_{-1}^{1} P_l^m(x) dx \, Y_{l,m}(\theta,\varphi)$$